Module 2

(Manual Testing)

- What is Exploratory Testing?

Exploratory testing is a software testing approach in which testers simultaneously learn the application, design test cases, and execute tests without predefined scripts.

- What is traceability matrix?

A Traceability Matrix is a document that maps and tracks requirements with test cases to ensure that all requirements are covered and tested.

- What is Boundary value testing?

Boundary Value Testing is a software testing technique in which test cases are designed to check values at the edges (boundaries) of input ranges, as errors often occur at these limits.

- What is Equivalence partitioning testing?

Equivalence Partitioning is a software testing technique in which input data is divided into groups (partitions) where each group is expected to behave similarly, and only one value from each group is tested.

- What is Integration testing?

Testing performed to expose defects in the interfaces and in the interactions between integrated components or systems

Integration Testing is a level of these of software testing process where individual unit share combined and tested as a group.

- What determines the level of risk?

A factor that could result in future negative consequences; usually expressed as **impact** and **likelihood** When testing does find defects, the Quality of the software system increases when those defects are fixed

- What is Alpha testing?

Alpha testing is a type of software testing performed by the internal team (developers or testers) at the developer's site before releasing the product to external users.

- What is beta testing?

Beta testing is a type of software testing performed by real users outside the organization in a real environment before the final release of the product.

- What is component testing?

Component testing is a type of software testing in which individual parts or components of a software application are tested independently to verify that each part works correctly.

- What is functional system testing?

A requirement that specifies a function that a system or system component must perform.

- What is Non-Functional Testing?

Testing the attributes of a component or system that do not relate to functionality, e.g. reliability, efficiency, usability, interoperability, maintainability and portability

- What is GUI Testing?

GUI (Graphical User Interface) testing is a type of software testing that verifies the visual elements of an application, such as buttons, menus, icons, and layouts, to ensure they work correctly and are user-friendly.

- What is Adhoc testing?

The Error guessing is a technique where the experienced and good testers are encouraged to think of situations in which the software may not be able to cope.

- What is load testing?

Load testing is a type of software testing that checks how a system performs under expected normal and peak user loads to ensure it can handle the anticipated traffic without performance issues.

- What is stress Testing?

Stress testing is a type of software testing that evaluates how a system performs under extreme or beyondnormal conditions, such as heavy load, high traffic, or limited resources, to check its stability and reliability.

- What is white box testing and list the types of white box testing?

Testing based on an analysis of the internal structure of the component or system.

Structure-based testing technique is also known as „white-box" or „glass-box" testing technique because here the testers require knowledge of how the software is implemented, how it works.

Types:

1. statement coverage
2. Decision coverage
3. Condition coverage

- What is black box testing? What are the different black box testing techniques?

Black box testing is a software testing technique where the tester evaluates the functionality of an application ***without knowing the internal code or structure***. The focus is on inputs and expected outputs.

1. equivalence partitioning
2. boundary value analysis
3. decision table
4. state transition
5. use case

- ## Mention what are the categories of defects?

1. Functional Defects
2. User interface Defects
3. Performance Defects
4. Security Defects
5. Compatibility Defects
6. Usability Defects

- ## Mention what big bang testing is?

Big Bang Testing is an integration testing approach where all the modules of a software application are combined together at once and tested as a complete system

- ## What is the purpose of exit criteria?

Purpose of exit criteria is to define when we STOP testing either at the:

End of all testing – i.e. product Go Live

End of phase of testing (e.g. hand over from System Test to UAT)

- ## When should "Regression Testing" be performed?

Regression testing should be performed whenever changes are made to the software to ensure that existing functionality still works correctly.

- ## What is 7 key principles? Explain in detail?

1. Testing shows Presence of Defects
   - Testing can find bugs, but cannot prove that there are no bugs.
   - Example: Even if app works, hidden bugs may still exist.

2. Exhaustive testing is impossible
   - You cannot test everything (all inputs, all cases).
   - So we test based on priority and risk.

3. Early testing saves time and cost
   - Start testing in early stages (requirement/design).
   - Finding bugs early is cheaper.

4. Defect clustering
   - Most defects are found in a small part of the system.

- Example: 80% bugs may be in 20% modules.

5. Pesticide paradox
    - Same test cases again and again will not find new bugs.
    - You must update and improve test cases.
6. Testing is context dependent
    - Testing depends on application type.
    - Example: Banking app needs more security testing than a game.
7. Absence of errors is a fallacy
    - Even if no bugs are found, software may still be useless if it does not meet user needs.

• Difference between QA v/s QC v/s Tester

Testing	QC	QA
Subset of QC	Quality Control	Quality Assurance
Focus on test execution	Focus on product	Focus on process
Done during development or after development	Done after development	Done before development
Actual Testing	Finds Defect	Prevents Defect

• Difference between Smoke and Sanity?

Smoke	Sanity
1. To check the critical functionality of new build	1. To Test the after defects fix and new functionality
2. It checks the stability of software	2.It checks the Rationality of software
3.General health checkup	3. specialized health check up

• Difference between verification and Validation?

Verification	Validation
1. Static Testing ("Review" of document)	1. Dynamic Testing ("Live Testing" of actual software)
2. Are you building the product right	2. Are you building the right product
3. Review, walkthrough, inspection	3. Testing

- Explain types of Performance testing? \

Load Testing

Checks system behaviour under expected number of users.

Example: 100 users using a website at the same time.

Stress Testing

Tests system beyond its limits to see breaking point.

Example: Increasing users until system crashes.

Spike Testing

Tests system response to sudden increase/decrease in users.

Example: 100 to1000 users come

Volume Testing

Checks performance with large amount of data.

Example: Huge database records.

Endurance Testing (Soak Testing)

Tests system for long duration.

Example: System runs for 24–48 hours continuously.

Scalability Testing

Checks if system can handle growth (users/data).

Example: Can system support more users if increased?

Capacity Testing

Determines maximum user system can handle efficiently.

- What is Error, Defect, Bug and failure?

Error: A mistake in coding is called error, error found by tester is called defect, defect accepted by development team then it is called bug, build does not meet the requirements then it is failure
Defect: Commonly refers to several troubles with the software products, with its external behaviour or with its internal features.
Bug: A fault in a program which causes the program to perform in an unintended or unanticipated manner. See: anomaly, defect, error, exception, and fault. Bug is terminology of Tester.
Failure: The inability of a system or component to perform its required functions within specified performance requirements. See: bug, crash, exception, and fault.

- Difference between Priority and Severity?

Basics	Severity	Priority
Definition	It refers to the impact of a defect on the system	It refers to the urgency of fixing a defect
Meaning	How serious the bug is	How quickly it should be fixed
Decided by	Tester	Manager / Client
Focus	Technical impact	Business impact
Example	Application crash	Spelling mistake on homepage
Type	High, Medium, Low	High, Medium, Low

- What is Bug Life Cycle?

Bug Life Cycle is a process in software testing where a defect goes through different stages like New, Assigned, Open, Fixed, Retest, Verified, and Closed until it is resolved.

- Explain the difference between Functional testing and Non-Functional testing?

Functional Testing	Non-Functional Testing
Tests *what the system does*	Tests how *the system works*
Focus on features and functions	Focus on features and functions
Based on functional requirements	Based on non-functional requirements

- To create HLR & Test-Cases of

(1) (Instagram, Facebook) first page and chat functionality
(2) Facebook Login Page : https://www.facebook.com/

- What is the difference between the STLC (Software Testing Life Cycle) and SDLC (Software Development Life Cycle)

SDLC (Software Development Life Cycle)	STLC (Software Testing Life Cycle)
it is the process of developing software	it is the process of Testing software
Focuses on building the application	Focuses on checking quality of application
Performed by developers, designers, business analysts	Performed by testers / QA team
Main goal is to create working software	Main goal is to find bugs and ensure quality

- What is the difference between test scenarios, test cases, and test script?

Test Scenario	**Test case**	**Test Script**

High-level idea of what to test	Detailed steps to test a functionality	Automation code to execute test cases
Very short and simple	Detailed and structured	Written in programming language
No steps included	Includes steps, input, expected result	Includes commands, logic, scripts
Used in manual testing planning	Used in manual testing execution	Used in automation testing

- Explain what Test Plan is? What is the information that should be covered

A Test Plan is a document that describes how testing will be done for a project. It defines the scope, strategy, resources, schedule, and activities of testing.

- **Objective** – Goal of testing
- **Scope** – What to test and what not to test
- **Test Strategy** – Type of testing to be used
- **Test Environment** – Tools and system details
- **Roles & Responsibilities** – Who will do what
- **Schedule** – Start and end time of testing
- **Entry & Exit Criteria** – When to start and stop testing
- **Risks** – Possible problems and solutions

- What is priority?
  Priority defines how quickly a defect (bug) should be fixed.
- What is severity?
  Severity defines how serious a defect (bug) is and how much it affects the system
- Bug categories are…
  - **Functional Bug** – Feature not working correctly
  - **UI Bug** – Issue in design, layout, or appearance
  - **Performance Bug** – Slow speed or response time
  - **Security Bug** – Data or system not secure
  - **Compatibility Bug** – Not working on different devices/browsers
  - **Usability Bug** – Difficult to use for users
  - **Logic Bug** – Wrong calculation or incorrect logic

- Advantage of Bugzila.
  - **Open Source** – Free to use
  - **Easy Bug Tracking** – Manage and track defects easily
  - **Email Notifications** – Get updates on bug status
  - **Customizable** – Can be modified as per project needs
  - **User Management** – Control access for different users
  - **Report Generation** – Create bug reports easily

• **Integration Support** – Works with other tools

- What are the different Methodologies in Agile Development Model?

  • **Scrum** – Work is divided into small sprints
  • **Kanban** – Work is managed using boards and continuous flow
  • **Extreme Programming (XP)** – Focus on coding quality and frequent releases
  • **Lean** – Reduces waste and improves efficiency
  • **Crystal** – Focus on team communication and simplicity
  • **Feature Driven Development (FDD)** – Develop features step by step

- Explain the difference between Authorization and Authentication in Web testing. What are the common problems faced in Web testing?

Authorization	Authentication
Verifies user identity	Checks user permissions
Answers: *Who are you?*	Answers: *What can you access?*
Done using login (username & password)	Done after login
Example: User login	Example: Access to admin page

Common Problems in Web Testing:

- **Browser Compatibility Issues** – Site not working on all browsers
- **Performance Issues** – Slow loading website
- **Security Issues** – Data not (not secure)
- **Broken Links** – Links not working
- **UI Issues** – Design/layout problems
- **Session Issues** – Login/logout problems
- **Data Validation Issues** – Wrong input accepted
- **Network Issues** – Internet-related problems